



Dr. Abderraouf Maoudj

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🌐 Citations +1700 📈 h-index 13 📊 i10-index 15

About

Dr. Abderraouf Maoudj is a senior robotics researcher specializing in multi-robot learning and coordination, collaborative manipulation, and AI-driven autonomous systems, with a strong focus on applications for Industry 4.0 environments. With more than a decade of academic and industrial R&D experience across Denmark, Algeria, and Saudi Arabia, he designs algorithms that move beyond theory into large-scale deployments in dynamic warehouses, factories, and smart manufacturing settings.

Currently a Post-Doctoral Research Fellow at the Interdisciplinary Research Center for Intelligent Manufacturing & Robotics (IRC-IMR), KFUPM, he leads research on Next-Generation Robotic Manipulation and Learning Systems, where robots act as adaptive apprentices rather than rigid automation. His current agenda integrates vision–language–action (VLA) models, world-model-based reinforcement learning, and collaborative robotics to create systems that can be instructed, taught, and coordinated rather than manually programmed. Concretely, his research aims to:

- Learn tool-usage and combine skills by observing human actions and inferring intent.
- Learn new manipulation skills from minimal human or video demonstrations through VLA models grounded in real industrial environments.
- Adapt and refine control policies entirely offline using learned world models, reducing on-robot experimentation and downtime.
- Enable fleets of cobots and mobile robots to coordinate as a distributed team when tasks exceed the capabilities or workspace of a single robot.

Prior to joining IRC-IMR, Dr. Abderraouf completed a postdoctoral appointment at the SDU Biorobotics section of the University of Southern Denmark, where he developed decentralized AI-based algorithms for swarm coordination in large e-commerce warehouses, lifelong multi-agent path finding, and capacitated multi-AGV scheduling. Earlier, as a Researcher at Algeria's Center for the Development of Advanced Technologies (CDTA), he contributed to designing Industry 4.0 platforms that integrate collaborative arms, AGV fleets, PLC-based automation, and intelligent scheduling into coherent cyber–physical assembly cells. Across these roles, his research has consistently focused on enabling large teams of robots to make local decisions that lead to globally efficient, robust behavior in dynamic, uncertain environments.

Dr. Abderraouf has led and contributed to projects with industrial partners such as Siemens, Enable Robotics, Mobile Industrial Robots (MiR), and Sonacome, delivering AI-enabled solutions for manufacturing, logistics, and inspection. These collaborations ensure that his methods are not only publishable but also deployable, bridging the gap between advanced robotic learning algorithms and production-ready systems for next-generation industrial automation.

Education

2013 – 2018	Ph.D. Process Control & Robotics , University of Sciences and Technology Houari Boumediene (USTHB). <i>Thesis:</i> Remote Control of a Heterogeneous Multi-Robot System: application to the RobuTER/ULM and AGV-M6 platforms.
2010 – 2012	M.Sc. Electronics & Control , Mohamed Seddik Ben Yahia University, Jijel. <i>Thesis:</i> Design and Realisation of a Mobile Robot.

2007 – 2010

B.Sc. Electrical Engineering, Mohamed Seddik Ben Yahia University, Jijel.
Capstone: Development of a Card Reader.

Professional Experience: Research, Teaching & Supervision

2023–Present

Post-Doctoral Researcher (research & teaching), Interdisciplinary Research Center for Intelligent Manufacturing & Robotics (IRC-IMR), KFUPM.

Research: Leading projects on adaptive control and learning for next-generation robotic manipulation systems; collaborative robotics frameworks with closed-loop control; robust AI-based skill learning for human–robot collaboration in smart manufacturing; and AI-based methods for high-throughput coordination of autonomous mobile robot fleets in dense warehouses and fulfillment centers supporting KSA’s e-commerce and logistics sectors.

Teaching & supervision: Co-designed and delivered graduate course. Supervise Master’s students and research interns on topics including multi-robot motion planning, decentralized coordination, and learning-based control for industrial robotics, with several projects progressing to publications, industrial demonstrations, or technology prototypes.

2021–2023

Post-Doctoral Researcher (research & supervision), SDU Biorobotics, Denmark.

Research: Focused on AI-based distributed multi-robot coordination for large-scale e-commerce logistics, including decentralized multi-agent path finding, capacitated multi-AGV scheduling, and swarm robotics in dynamic warehouse environments.

Supervision: Contributed to graduate mentoring in robotics and intelligent systems, including guest lectures and project supervision in courses related to multi-robot systems. Supervised and co-supervised Master’s projects on decentralized multi-agent path finding, emphasizing reproducible experimentation, benchmarking, and open-source software.

2018–2020

Research Professor (research & teaching), Robotics Division, CDTA, Algeria.

Research: Conducted research and development on robotics and automation solutions for smart manufacturing, with emphasis on scalable integration of collaborative robots, AGV fleets, PLC-based automation, and intelligent scheduling into Industry 4.0 platforms.

Teaching & supervision: Delivered core engineering courses for undergraduate and Master’s students, aligning course outcomes with modern industrial requirements. Supervised multiple Master’s theses and final-year projects in robotic manipulation, industrial automation, and autonomous navigation.

2014–2016

Robotics Research Engineer, Robotics Division, CDTA, Algeria.

Research & development : Integration of robotics in intelligent manufacturing cells, including implementation of navigation, perception, and control modules for mobile robots and collaborative arms in pilot Industry 4.0 environments.

Mentoring: Supported laboratory-based teaching and training activities for graduate students in control and automation.

Summary of Teaching & Supervision Across Institutions: 2014–Present

Teaching and mentorship are central pillars of my academic career. I design and deliver learning experiences that bridge foundational theory with implementation, ensuring that students acquire the analytical, problem-solving, and engineering skills required in robotics, control, automation, and intelligent systems. My pedagogy emphasizes conceptual understanding, systems thinking, and hands-on application through simulation, laboratory work, and design-based projects.

Courses Delivered (Undergraduate and Graduate):

- **Linear Control Systems** (UG): Modeling of dynamic systems, Laplace transforms, time- and frequency-domain responses, and controller synthesis, with extensive MATLAB/Simulink-based validation and lab experiments.
- **Introduction to Robotic Systems and ROS** (UG/Grad): Mobile and manipulator kinematics and dynamics, sensing and actuation, control architectures, and ROS 1/2 fundamentals (Navigation Stack, MoveIt, real-time interfacing with physical robots).
- **Automation and PLCs** (UG): Industrial automation principles; Siemens TIA Portal (Ladder, Grafset); actuator–sensor integration; and PLC-based design of discrete-event control systems for manufacturing.
- **Applied Reinforcement Learning for Robotic Manipulation** (Grad): Manipulator kinematics and control; Q-learning, DQN, and SAC; implementation of learning-based controllers for robotic arms in simulation and on hardware, with coding-intensive assignments and mini-projects.
- **Embedded Systems for Automotive Engineering** (UG): Fundamentals of automotive embedded systems; sensing and actuation for automotive applications; introduction to AUTOSAR and vehicle control units; real-time implementation constraints.

Student Supervision:

Supervised and co-supervised PhD students and Master’s students, working on robotic manipulation, decentralized multi-robot systems, autonomous navigation, Industry 4.0 integration, and reinforcement learning for control. Many of these projects led to scientific publications, simulation-to-reality deployments in lab testbeds, or prototype implementations in collaboration with industrial partners.

Seminars and Invited Lectures

1. **Decentralized Multi-Robot Motion Planning and Coordination for Industrial Applications**
Invited seminar, University of Tartu, Estonia, September 2024. ([Public link](#))
2. **Reinforcement Learning in Robotics: From Mathematical Foundations to Real-World Applications**
Invited lecture, Department of Electrical Engineering, King Fahd University of Petroleum & Minerals, Saudi Arabia, November 2025.

Workshops, Exhibitions and Outreach

- **Fourth Annual Research Workshop at DTV, KFUPM**, 19–20 August 2024 ([Workshop link](#)).
Co-organized the workshop and presented a seminar entitled *Multi-Robot Motion Planning and Coordination in Industrial Environments*. The event facilitated collaboration on future projects and grants aligned with KFUPM’s grand challenges, with sessions in manufacturing, robotics, UAVs, automation, machine learning, and AI.
- **2024 KFUPM Research Forum: New Research Identity for a Better KSA and World**, 2–3 October 2024 ([Exhibition link](#)).
Represented the IRC-IMR Robotics activities, presenting our research strategy aligned with Vision 2030 and demonstrating collaborative robotics and multi-robot systems for Industry 4.0 to industrial partners.

Selected R&D Projects

- Ongoing Project 1 **Next-Generation Robotic Manipulation and Learning Systems for Saudi Industry 4.0** — Post-doctoral Researcher, IRC-IMR, KFUPM.
This research aims to develop and demonstrate next-generation robotic manipulation systems that fundamentally transform how robots acquire and execute skills. My vision is to enable robots to understand human intent through natural communication, take plain instructions, learn tasks from minimal demonstrations, and coordinate multi-robot teams when tasks exceed individual capabilities. The work combines generative reinforcement learning, large vision–language–action models for grounding instructions, world models for safe offline skill refinement, and distributed intelligence for collaborative robot coordination.
- Ongoing Project 2 **Resilient Legged Robotics: Decentralized RL–CPG Locomotion Under Progressive Hardware Malfunctions** — Post-doctoral Researcher, IRC-IMR, KFUPM.
This ongoing project develops and validates a failure-adaptive locomotion stack for multi-legged robots operating under progressive actuator and joint degradations. The core direction is to combine bio-inspired CPG limit-cycle dynamics with reinforcement learning that modulates coordination parameters online (adaptation), enabling graceful performance degradation without gait libraries or controller switching. Key outputs include:
- **Decentralized RL–CPG architecture:** Designed a leg-local Hopf-oscillator network with an RL modulation layer that reallocates stepping amplitude/timing when subsets of legs become unreliable.
 - **Fault-as-training protocol:** Implemented randomized multi-leg fault injection and curriculum training in MuJoCo to learn a single policy covering many leg-availability patterns, including protective *survival* behaviors when static locomotion becomes infeasible.
 - **Sim-to-real validation on hexapod robot:** Deployed the learned controller zero-shot on a physical 18-DoF hexapod and evaluated sequential malfunctions with quantitative stability and recovery-time measures (manuscript under Review in *IEEE Transactions on Robotics*).

2023 – 2025

Advancing Robotics for Industry 4.0: Integrating Collaborative Robot systems to Enhance Industrial tasks ([Project website](#)) — Post-doc, IRC-IMR, KFUPM.

Led Project Grant No. INMR2403, which redefines how collaborative robots learn, share, and execute skills in various dynamic production settings. The aim is to build a scalable collaborative robotics network that supports industrial environments with a foundation of shared and adaptable skills, in alignment with Saudi Arabia's Industrial Vision 2030. Contributions include:

- **Multi-cobot coordination and motion control:** Developed several AI-driven methods, including multi-agent deep reinforcement learning, imitation learning, and transformer-based inter-arm attention, for learning decentralized multi-arm motion planning. New videos of my work are available on our lab's channel: [video](#)
- **Swarm Robotics Coordination in Smart Manufacturing:** Addressed large-scale warehouse automation challenges in e-commerce, developing AI-based coordination mechanisms for fleets of mobile robots handling storage, transport, and shipment preparation. These algorithms improve efficiency in logistics operations and advance the foundations of smart manufacturing in line with Saudi Arabia's Industry 4.0 initiative.

2021 – 2023

Swarm Robotics for Industry 4.0 ([Project website](#)) — Post-doc Researcher, Denmark.

The Fourth Industrial Revolution transforms manufacturing through robotics, IoT, AI, and cloud computing. This project studies decentralized coordination between mobile robotics in I4.0 manufacturing and warehouse scenarios to achieve higher levels of autonomy, scalability, flexibility, and robustness than current solutions offer. In this project, we developed AI-based algorithms for multi-robot planning and coordination, enabling groups of autonomous agents to distribute tasks, adapt to changing conditions, and collaborate toward shared goals. Key outcomes:

- *Capacitated multi-agent control/scheduling framework:* Aiming to boost efficiency in logistics, warehouses, and transportation sectors, I designed and implemented a fully decentralized capacitated multi-agent control and scheduling framework specifically targeting logistics optimization. This system enables each autonomous guided vehicle (AGV) to reason locally while achieving fleet-level optimal solutions under capacity and compatibility constraints.
- *Swarm robots motion coordination in dynamic industrial environments and fulfillment centers:* Developed fast and robust decentralized planners coordinating up to thousands of robots in large dynamic warehouses. These research works improve order fulfillment efficiency, enhance operational efficiency, reduce costs, and strengthen global supply chain resilience.
- *Open-source contributions:* Released the **ANTS** planner and benchmark suite for research and teaching ([GitHub](#), [Benchmark dataset for lifelong MAPF](#), and [video](#)), now used in academia.

- 2014 – 2016 **Remote Control of Heterogeneous Robots** — Robotics Engineer|PhD student, CDTA.
As part of my doctoral research, this project involved:
- Developing a distributed multi-agent system for controlling industrial robot fleets.
 - Implemented DRL-driven path optimization and fuzzy-logic controllers for mobile manipulators.
 - Developing a fault-tolerant leader–follower control system for swarm mobile robots that maintains formation integrity under single-robot failures.
- 2016 – 2020 **Industry 4.0 Platform Development** — Robotics Researcher, CDTA.
This project entailed designing and implementing advanced algorithms and control systems for Industry 4.0 platforms, seamlessly integrating robotic innovations with smart automation principles. Key responsibilities included:
- Integrated KUKA LBR iiwa with Java/ROS middleware for collaborative assembly.
 - Implemented an RFID-based part-tracking system linked to Siemens PLCs, boosting inventory accuracy.
 - Upgraded the facility’s AGV fleet with a deep reinforcement learning navigation system to align with the control architecture of the I4.0 platform.
- 2016 – 2018 **Aquatic Boat Robot for Bathymetry** — Robotics Researcher, CDTA.
This project aimed to provide a secure solution for bathymetric measurement using a remotely controlled boat robot for the Algerian National Hydrological Agency. My key role in this project focused on the electronic realization and the control of the robot.

Skills

Robotics: Python/C++, ROS 1/2, KUKA LBR iiwa, MiR, Enable Robotics MM, Meca500.

Control Electronics: Sensor conditioning; STM32/Arduino; Siemens TIA-Portal; real-time PLC control.

Software: Python, C/C++, Java, C#, MATLAB, R; Git, Docker, Linux.

List of Publications

For a complete list of my publications, refer to my Google Scholar profile: [Scholar profile](#)

References

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